

Water & Wastewater I

FE Environmental — Day 7

Study set • 2026-06-23 • 21 questions • every numeric answer recomputed in Python

Handbook fidelity. All formulas cross-checked against NCEES FE Reference Handbook v10.6 (Environmental Engineering, pp. 327–353). Confirmed Handbook formulas: BOD exertion $BOD_t = L_o(1 - e^{-kt})$ (p.327); mass loading M (lb/day) $= C$ (mg/L) $\times Q$ (MGD) $\times 8.34$ (p.327); CMFR first-order $C_t = C_0/(1 + k\theta)$ (p.331 reactor table); clarifier overflow rate $v_o = Q/A$ and design-criteria table (pp.344–345); Stokes' Law $v_t = g(\rho_p - \rho_f)d^2/(18\mu)$ (p.346); CT disinfection $CT_{calc} = C \times t_{10}$ with baffling-factor table (pp.351–352); lime-soda softening reactions 1–7 (p.347); alum coagulation reaction (p.347); BOD test $BOD = (D_1 - D_2)/P$ (p.339); kinetic temperature correction $k_T = k_{20}\theta^{T-20}$, $\theta = 1.056$ for BOD at 21–30°C (p.328); activated sludge $X_A = \theta_c Y(S_0 - S_e)/[\theta(1 + k_d\theta_c)]$ and SVI (p.339); biotower fixed-film $S_e/S_0 = e^{-kD/q^n}$ (p.340); sludge volume $V = W_s/[(s/100)\gamma S]$ (p.338); aerobic digestion SRT table (p.341). Hardness conversion uses the Common Radicals equivalent-weight table (p.348): $EW(\text{Ca}^{2+}) = 20.0$, $EW(\text{Mg}^{2+}) = 12.2$, $EW(\text{CaCO}_3) = 50.0$. Per capita values (100 gpcd, 0.20 lb BOD₅/person/day) in Q20 are SUPPLEMENTAL (not in Handbook v10.6) and are stated in the question stem (tagged in-stem).

Q1. [D7-Q01 • MCQ • Total hardness as CaCO₃]

A water sample contains $\text{Ca}^{2+} = 60$ mg/L and $\text{Mg}^{2+} = 18$ mg/L. Using the equivalent weights from the FE Handbook (EW of $\text{Ca}^{2+} = 20.0$ g/eq, $\text{Mg}^{2+} = 12.2$ g/eq, $\text{CaCO}_3 = 50.0$ g/eq), the total hardness expressed as mg/L CaCO₃ is most nearly:

Relevant: $\text{Hardness as CaCO}_3 = C_{ion} \times \frac{EW_{\text{CaCO}_3}}{EW_{ion}}$ Total hardness = Ca hardness + Mg hardness

- (A) 74 mg/L
- (B) 150 mg/L
- (C) 187 mg/L
- (D) **224 mg/L** ✓

Answer: (D)

Ca hardness $= 60 \times (50.0/20.0) = 60 \times 2.50 = 150.0$ mg/L as CaCO₃. Mg hardness $= 18 \times (50.0/12.2) = 18 \times 4.098 = 73.8$ mg/L as CaCO₃. Total $= 150.0 + 73.8 = 223.8 \approx 224$ mg/L as CaCO₃. Option (A) is Mg hardness only; (B) is Ca hardness only; (C) uses the Mg atomic weight 24.3 as the EW instead of dividing by 2.

Handbook: Water and Wastewater — Common Radicals in Water (equivalent weight table) • Ctrl-F: Common Radicals; Equivalent Weight; Hardness

Q2. [D7-Q02 • MCQ • BOD₅ test — unseeded dilution method]

A BOD₅ test is run on unseeded, diluted wastewater. The initial dissolved-oxygen (DO) of the diluted sample is $D_1 = 8.1$ mg/L; after 5-day incubation at 20°C, $D_2 = 2.4$ mg/L. The dilution fraction is $P = 0.05$ (5 mL sample in 100 mL total). The BOD₅ of the wastewater is most nearly:

Relevant: $BOD \text{ (mg/L)} = \frac{D_1 - D_2}{P}$

- (A) 5.7 mg/L
- (B) 57 mg/L
- (C) **114 mg/L** ✓
- (D) 162 mg/L

Answer: (C)

$BOD_5 = (D_1 - D_2)/P = (8.1 - 2.4)/0.05 = 5.7/0.05 = 114$ mg/L. Option (A) is $(D_1 - D_2)$ without dividing by P ; (B) uses $P = 0.10$ by mistake; (D) uses D_1 alone ($8.1/0.05 = 162$).

Handbook: Environmental Engineering — BOD Test Solution and Seeding Procedures • Ctrl-F: BOD Test; Dilution

Q3. [D7-Q03 • MCQ • Mass loading conversion — lb/day]

A municipal wastewater treatment plant receives a flow of $Q = 2.5$ MGD with an influent BOD₅ concentration of 220 mg/L. The daily BOD₅ mass loading M in lb/day is most nearly:

Relevant: $M \text{ (lb/day)} = C \text{ (mg/L)} \times Q \text{ (MGD)} \times 8.34 \text{ [lb} \cdot \text{L}/(\text{mg} \cdot \text{MG})]$

- (A) 550 lb/day
- (B) 2,294 lb/day
- (C) **4,587 lb/day** ✓
- (D) 9,174 lb/day

Answer: (C)

$M = 220 \times 2.5 \times 8.34 = 4,587$ lb/day. Option (A) = $C \times Q = 550$ (missing the 8.34 unit-conversion factor); (B) = half the correct value; (D) = double the correct value.

Handbook: Environmental Engineering — Mass Balance / Flow Rate • Ctrl-F: lb/day; MGD; 8.34

Q4. [D7-Q04 • MCQ • CMFR (CSTR) first-order effluent concentration]

A completely mixed flow reactor (CMFR/CSTR) treats an industrial leachate. Influent concentration $C_0 = 250$ mg/L, first-order decay constant $k = 0.20$ day⁻¹, hydraulic residence time $\theta = 3$ days. The steady-state effluent concentration is most nearly:

Relevant: $C_t = \frac{C_0}{1 + k\theta}$ (CMFR, first-order) $C_t = C_0 e^{-k\theta}$ (PFR or batch, first-order)

- (A) 92 mg/L
- (B) 100 mg/L
- (C) 137 mg/L
- (D) **156 mg/L** ✓

Answer: (D)

For a CMFR (ideally mixed reactor) with first-order decay: $C_t = C_0/(1+k\theta) = 250/(1+0.20 \times 3) = 250/1.60 = 156$ mg/L. Option (A) is a batch/PFR result at $t = 5$ days ($250e^{-1.0}$); (B) uses a linear approximation $C_0(1 - k\theta) = 100$; (C) is the PFR/batch result at $\theta = 3$ days ($250e^{-0.6} = 137$). The CMFR is always less efficient than a PFR for positive-order reactions, so the CMFR effluent (156 mg/L) is higher than the PFR effluent (137 mg/L).

Handbook: Environmental Engineering — Steady-State Reactor Parameters (CMFR, first-order) • Ctrl-F: CMFR; Steady-State; First Order

Q5. [D7-Q05 • MCQ • Clarifier surface overflow rate]

A circular primary clarifier has a diameter of 80 ft and receives a flow of $Q = 4.0$ MGD. The surface overflow rate (SOR) in gpd/ft² is most nearly:

Relevant: $v_o = \frac{Q}{A_{surface}}$, $A_{circle} = \frac{\pi D^2}{4}$

- (A) 200 gpd/ft²
- (B) 500 gpd/ft²
- (C) **796 gpd/ft²** ✓
- (D) 1,200 gpd/ft²

Answer: (C)

$A = \pi D^2/4 = \pi(80)^2/4 = 5,027$ ft². $Q = 4,000,000$ gpd. $v_o = 4,000,000/5,027 = 796$ gpd/ft². This is within the Handbook primary-clarifier design range of 800–1,200 gpd/ft² (average). Option (A) uses πD^2 without the $\div 4$ factor, giving $A = 20,106$ ft² and $SOR \approx 199$ gpd/ft².

Handbook: Environmental Engineering — Clarifier (Overflow Rate; Design Criteria table) • Ctrl-F: Overflow Rate; Clarifier; Primary

Q6. [D7-Q06 • MCQ • Hydraulic residence time in a settling tank]

A rectangular settling basin has a volume $V = 600 \text{ m}^3$ and receives a flow $Q = 1\{,\}200 \text{ m}^3/\text{day}$. The hydraulic residence time θ in hours is:

Relevant: $\theta = \frac{V}{Q}$

- (A) 0.5 hr
- (B) 6 hr
- (C) **12 hr** ✓
- (D) 24 hr

Answer: (C)

$\theta = V/Q = 600/1,200 = 0.50 \text{ day} = 0.50 \times 24 = 12 \text{ hr}$. The Handbook design table shows primary clarifiers typically have HRT of 2 hr, so 12 hr would indicate over-design or a different unit process.

Handbook: Environmental Engineering — Clarifier (Hydraulic Residence Time) • Ctrl-F: Hydraulic Residence Time; Detention Time

Q7. [D7-Q07 • MCQ • Stokes' Law — particle settling velocity]

Estimate the terminal settling velocity of a spherical silica particle (diameter $d = 0.10 \text{ mm}$, particle density $\rho_p = 2\{,\}650 \text{ kg/m}^3$) settling in water at 20°C ($\rho_f = 1\{,\}000 \text{ kg/m}^3$, $\mu = 1.0 \times 10^{-3} \text{ Pa}\cdot\text{s}$). Stokes' Law (laminar settling) applies. The terminal velocity v_t is most nearly:

Relevant: $v_t = \frac{g(\rho_p - \rho_f) d^2}{18\mu}$

- (A) $4.5 \times 10^{-4} \text{ m/s}$
- (B) $9.0 \times 10^{-4} \text{ m/s}$
- (C) $4.5 \times 10^{-3} \text{ m/s}$
- (D) **$9.0 \times 10^{-3} \text{ m/s}$** ✓

Answer: (D)

$d = 0.10 \text{ mm} = 1.0 \times 10^{-4} \text{ m}$; $d^2 = 1.0 \times 10^{-8} \text{ m}^2$. $v_t = \frac{9.81 \times (2,650 - 1,000) \times 10^{-8}}{18 \times 10^{-3}} = \frac{9.81 \times 1,650 \times 10^{-8}}{0.018} = \frac{1.619 \times 10^{-4}}{0.018} = 9.0 \times 10^{-3} \text{ m/s}$. Note: the exponent is 10^{-3} , not 10^{-4} — the most common error is a powers-of-ten mistake in unit conversion.

Handbook: Environmental Engineering — Settling Equations (Stokes' Law) • Ctrl-F: Stokes\2019 Law; Settling Velocity; Terminal Velocity

Q8. [D7-Q08 • Numeric • Primary clarifier — surface area design (USCS)]

A 2.0-MGD wastewater plant must size its primary clarifier. Using the Handbook average design overflow rate for primary clarifiers (800 gpd/ft²), determine the required plan-view surface area A (in ft²).

Relevant: $A = \frac{Q}{v_o}$

Numeric entry.

Answer: 2,500 ft²

$Q = 2.0 \text{ MGD} = 2,000,000 \text{ gpd}$. $A = Q/v_o = 2,000,000/800 = 2,500 \text{ ft}^2$. The Handbook design criteria table gives 800–1{,}200 gpd/ft² (average) for primary clarifiers; 800 gpd/ft² is the conservative (lower-end) design value.

Handbook: Environmental Engineering — Clarifier (Design Criteria for Sedimentation Basins table) • Ctrl-F: Primary Clarifier; Overflow Rate; Design Criteria

Q9. [D7-Q09 • MCQ • CT disinfection — contact-chamber design]

A chlorine contact chamber has a hydraulic residence time $\theta = 30 \text{ min}$ and a baffling factor $BF = 0.5$ (average baffling per the Handbook table). The chlorine residual measured at peak hourly flow is $C = 1.5 \text{ mg/L}$. The calculated CT value (CT_{aala}) in mg·min/L is most nearly:

Relevant: $CT_{\text{calc}} = C \times t_{10}$ $t_{10} \approx \theta \times BF$

- (A) 3.5 mg·min/L
- (B) 15 mg·min/L
- (C) **22.5 mg·min/L** ✓
- (D) 45 mg·min/L

Answer: (C)

$t_{10} = \theta \times BF = 30 \times 0.5 = 15 \text{ min}$. $CT_{\text{calc}} = C \times t_{10} = 1.5 \times 15 = 22.5 \text{ mg} \cdot \text{min/L}$. Option (B) is t_{10} alone (forgot to multiply by C); option (D) uses $BF = 1.0$ (perfect plug flow, $CT = 1.5 \times 30 = 45$). A CT of 22.5 mg·min/L provides substantial Giardia inactivation credit at typical temperatures.

Handbook: Environmental Engineering — Disinfection (CT; Baffling Factors table) • Ctrl-F: CT; Baffling Factor; Disinfection

Q10. [D7-Q10 • MCQ • Lime-soda softening — calcium carbonate hardness removal]

In lime-soda softening, which of the following reactions is used to remove calcium carbonate hardness from water?

- (A) $\text{CO}_2 + \text{Ca}(\text{OH})_2 \rightarrow \text{CaCO}_3(\text{s}) + \text{H}_2\text{O}$ [reaction 1 — CO_2 removal]
- (B) **$\text{Ca}(\text{HCO}_3)_2 + \text{Ca}(\text{OH})_2 \rightarrow 2\text{CaCO}_3(\text{s}) + 2\text{H}_2\text{O}$ [reaction 2 — calcium carbonate hardness] ✓**
- (C) $\text{CaSO}_4 + \text{Na}_2\text{CO}_3 \rightarrow \text{CaCO}_3(\text{s}) + 2\text{Na}^+ + \text{SO}_4^{2-}$ [reaction 3 — calcium non-carbonate hardness]
- (D) $\text{MgSO}_4 + \text{Ca}(\text{OH})_2 + \text{Na}_2\text{CO}_3 \rightarrow \text{CaCO}_3(\text{s}) + \text{Mg}(\text{OH})_2(\text{s}) + 2\text{Na}^+ + \text{SO}_4^{2-}$ [reaction 5 — Mg non-carbonate hardness]

Answer: (B)

Option (B) is Handbook Reaction 2: calcium carbonate (temporary) hardness is removed by adding lime $\text{Ca}(\text{OH})_2$ to water containing calcium bicarbonate, precipitating CaCO_3 and leaving only water. Option (A) removes dissolved CO_2 (not hardness). Option (C) removes calcium non-carbonate (permanent) hardness using soda ash Na_2CO_3 . Option (D) removes magnesium non-carbonate hardness using both lime and soda ash.

Handbook: Environmental Engineering — Lime-Soda Softening Equations (reactions 1–7) • Ctrl-F: Lime-Soda Softening; Calcium Carbonate Hardness

Q11. [D7-Q11 • MCQ • Alum coagulation — reaction products in alkaline water]

When aluminum sulfate (alum, $\text{Al}_2(\text{SO}_4)_3$) reacts with naturally alkaline water containing calcium bicarbonate $\text{Ca}(\text{HCO}_3)_2$, the reaction products are:

Relevant: $\text{Al}_2(\text{SO}_4)_3 + 3\text{Ca}(\text{HCO}_3)_2 \rightleftharpoons 2\text{Al}(\text{OH})_3 + 3\text{CaSO}_4 + 6\text{CO}_2$

- (A) $\text{Al}_2(\text{SO}_4)_3$ and $\text{Ca}(\text{HCO}_3)_2$ are regenerated — no net reaction
- (B) **$2\text{Al}(\text{OH})_3$ (settling floc), 3CaSO_4 , and 6CO_2 ✓**
- (C) $\text{Al}(\text{OH})_3$ and $\text{CaCO}_3(\text{s})$ only
- (D) $\text{Al}(\text{OH})_3$, CaCl_2 , and CO_2 [ferric-chloride coagulation products]

Answer: (B)

The Handbook coagulation reaction (reaction 1) gives products $2\text{Al}(\text{OH})_3$ (the gelatinous floc that enmeshes particles), 3CaSO_4 (remains in solution), and 6CO_2 (released). Option (C) is incorrect — CaCO_3 is a lime-soda softening product, not a coagulation product. Option (D) lists CaCl_2 , which would result from ferric chloride FeCl_3 reacting with $\text{Ca}(\text{HCO}_3)_2$, not from alum.

Handbook: Environmental Engineering — Coagulation Equations (reaction 1: alum in natural alkaline water) • Ctrl-F: Coagulation; Aluminum Sulfate; Alum

Q12. [D7-Q12 • MCQ • BOD exertion at 5 days]

A wastewater has an ultimate BOD (L_0) of 250 mg/L and a first-order BOD decay rate constant $k = 0.20 \text{ day}^{-1}$ (base e). The 5-day BOD (BOD_5) — the amount of oxygen demand exerted by day 5 — is most nearly:

Relevant: $BOD_t = L_o (1 - e^{-kt})$

- (A) 50 mg/L
- (B) 92 mg/L
- (C) **158 mg/L** ✓
- (D) 250 mg/L

Answer: (C)

$BOD_5 = 250 \times (1 - e^{-0.20 \times 5}) = 250 \times (1 - e^{-1.0}) = 250 \times (1 - 0.368) = 250 \times 0.632 = 158 \text{ mg/L}$. Option (B) $= 250 \times e^{-1} = 92$ is the BOD remaining (not yet exerted), the complement of the correct answer. Option (D) is the ultimate BOD L_0 itself, which is only approached asymptotically.

Handbook: Environmental Engineering — BOD Exertion (Microbial Kinetics) • Ctrl-F: BOD Exertion; Ultimate BOD; Microbial Kinetics

Q13. [D7-Q13 • Numeric • Ultimate BOD from a 5-day BOD measurement]

A laboratory measurement gives $BOD_5 = 185 \text{ mg/L}$ at 20°C with a BOD rate constant $k = 0.23 \text{ day}^{-1}$ (base e). Determine the ultimate BOD L_0 (in mg/L).

Relevant: $L_o = \frac{BOD_t}{1 - e^{-kt}}$

Numeric entry.

Answer: $\approx 271 \text{ mg/L}$

$L_o = \frac{BOD_5}{1 - e^{-k \times 5}} = \frac{185}{1 - e^{-0.23 \times 5}} = \frac{185}{1 - e^{-1.15}} \cdot e^{-1.15} = 0.3166$, so $1 - 0.3166 = 0.6834$. $L_o = 185/0.6834 = 270.7 \approx 271 \text{ mg/L}$.

Handbook: Environmental Engineering — BOD Exertion (Microbial Kinetics) • Ctrl-F: BOD Exertion; Ultimate BOD

Q14. [D7-Q14 • MCQ • BOD rate constant — kinetic temperature correction]

The BOD rate constant at 20°C is $k_{20} = 0.15 \text{ day}^{-1}$. Using the Handbook temperature correction for BOD in the range 21–30°C ($\theta = 1.056$), the corrected rate constant k at 25°C is most nearly:

Relevant: $k_T = k_{20} \theta^{T-20}$ $\theta = 1.056$ (BOD, $T = 21\text{--}30^\circ\text{C}$)

- (A) 0.150 day^{-1}
- (B) 0.178 day^{-1}
- (C) **0.197 day^{-1}** ✓
- (D) 0.212 day^{-1}

Answer: (C)

$k_{25} = 0.15 \times 1.056^{(25-20)} = 0.15 \times 1.056^5$. $1.056^5 = 1.3132$. $k_{25} = 0.15 \times 1.3132 = 0.197 \text{ day}^{-1}$. Option (A) = no correction applied; (B) uses bio-tower $\theta = 1.035$ ($0.15 \times 1.035^5 = 0.178$); (D) uses trickling-filter $\theta = 1.072$ ($0.15 \times 1.072^5 = 0.212$). The Handbook lists four separate θ values — choosing the correct one (1.056 for BOD at 21–30°C) is the key skill.

Handbook: Environmental Engineering — Kinetic Temperature Corrections • Ctrl-F: Temperature Correction; kT ; Theta

Q15. [D7-Q15 • Numeric • Activated sludge — MLSS concentration (SI)]

A conventional activated-sludge system treats municipal wastewater with the following design parameters: SRT $\theta_a = 10$ days, yield $Y = 0.60 \text{ kg VSS/kg BOD removed}$, influent BOD $S_0 = 200 \text{ mg/L}$, effluent BOD $S_e = 20 \text{ mg/L}$, endogenous decay $k_n = 0.05 \text{ day}^{-1}$, and hydraulic retention time $\theta = 8$ hr. Determine the mixed-liquor suspended solids (MLSS) concentration X_A in mg/L.

Relevant: $X_A = \frac{\theta_c Y (S_0 - S_e)}{\theta (1 + k_d \theta_c)}$

Numeric entry.

Answer: $\approx 2,160 \text{ mg/L}$

$\theta = 8 \text{ hr} = 8/24 = 0.3333 \text{ day}$. $S_0 - S_e = 200 - 20 = 180 \text{ mg/L}$. $X_A = \frac{10 \times 0.60 \times 180}{0.3333 \times (1 + 0.05 \times 10)} = \frac{1,080}{0.3333 \times 1.50} = \frac{1,080}{0.500} = 2,160 \text{ mg/L}$. This is in the Handbook typical range for conventional activated sludge (1,500–3,000 mg/L MLSS).

Handbook: Environmental Engineering — Activated Sludge (XA formula) • Ctrl-F: Activated Sludge; MLSS; SRT

Q16. [D7-Q16 • MCQ • Sludge Volume Index (SVI)]

A 30-minute settleability test on mixed-liquor from an activated-sludge aeration tank gives a settled-sludge volume of 300 mL/L. The MLSS concentration is 2,500 mg/L. The Sludge Volume Index (SVI) is most nearly:

Relevant: $SVI = \frac{\text{Sludge volume after settling (mL/L)} \times 1,000}{MLSS \text{ (mg/L)}}$

- (A) 0.12 mL/g
- (B) 8.3 mL/g
- (C) **120 mL/g** ✓
- (D) 830 mL/g

Answer: (C)

$SVI = (300 \times 1,000)/2,500 = 120$ mL/g. SVI ≤ 80 is excellent settling; 80–150 is acceptable; > 150 indicates poor settling (bulking). This sludge at $SVI = 120$ is settling satisfactorily. Option (A) is SV/MLSS without the $\times 1,000$ factor (unit mismatch); (B) is MLSS/SV (inverted); (D) is $2,500/3 = 833$ (confused units).

Handbook: Environmental Engineering — Activated Sludge (Sludge Volume Index SVI) • Ctrl-F: Sludge Volume Index; SVI; MLSS

Q17. [D7-Q17 • MCQ • Biotower — fixed-film effluent BOD (without recycle, SI)]

A biotower (plastic-media fixed-film reactor, no recycle) treats municipal wastewater with influent BOD $S_0 = 200$ mg/L. Treatability constant $k = 0.06 \text{ min}^{-1}$ (at 20°C), media depth $D = 5$ m, hydraulic loading $q = 0.25 \text{ m}^3/(\text{m}^2 \cdot \text{min})$, media coefficient $n = 0.5$ (modular plastic). The effluent BOD S_e is most nearly:

Relevant: $\frac{S_e}{S_0} = e^{-kD/q^n}$

- (A) 60 mg/L
- (B) **110 mg/L** ✓
- (C) 148 mg/L
- (D) 172 mg/L

Answer: (B)

$q^n = 0.25^{0.5} = 0.500$. $kD/q^n = (0.06 \times 5)/0.500 = 0.30/0.500 = 0.60$. $S_e = 200 \times e^{-0.60} = 200 \times 0.549 = 109.8 \approx 110$ mg/L. Option (A) uses $n = 1$ (forgot the media exponent): $q^1 = 0.25$, $kD/q = 1.2$, $S_e = 200e^{-1.2} = 60$ mg/L. Option (C) ignores q entirely: $S_e = 200e^{-kD} = 200e^{-0.3} = 148$ mg/L. The Handbook value $n = 0.5$ for modular plastic media is the critical parameter.

Handbook: Environmental Engineering — Biotower / Fixed-Film Equation without Recycle • Ctrl-F: Biotower; Fixed-Film; Recycle

Q18. [D7-Q18 • MCQ • Sludge volume from dry solids mass (SI)]

A wastewater treatment plant produces 1{,}500 kg/day of dry sludge solids. The sludge has a solids content of $s = 4\%$ (by mass) and a specific gravity $S = 1.02$. Using the Handbook formula for sludge volume with γ (water density) = 1{,}000 kg/m³, the wet sludge volume produced per day is most nearly:

Relevant: $V = \frac{W_s}{(s/100) \gamma S}$

- (A) 14.7 m³/day
- (B) **36.8 m³/day** ✓
- (C) 73.5 m³/day
- (D) 294 m³/day

Answer: (B)

$V = \frac{W_s}{(s/100) \gamma S} = \frac{1,500}{(4/100) \times 1,000 \times 1.02} = \frac{1,500}{40.8} = 36.8 \text{ m}^3/\text{day}$. Option (A) uses $s = 10\%$: $1,500/(0.10 \times 1,020) = 14.7 \text{ m}^3/\text{day}$. Option (C) uses $s = 2\%$: $1,500/(0.02 \times 1,020) = 73.5 \text{ m}^3/\text{day}$. Option (D) uses $s = 0.5\%$ (extreme dilution). This formula converts dry-weight production to wet-sludge volume for design of thickening, storage, and hauling.

Handbook: Environmental Engineering — Specific Gravity for a Solids Slurry; Wastewater Treatment and Technologies
• Ctrl-F: Sludge Volume; Solids Content; Specific Gravity

Q19. [D7-Q19 • MCQ • Aerobic digestion — minimum SRT at 15°C]

According to the design criteria for aerobic digesters in the NCEES FE Reference Handbook, the minimum solids residence time (SRT) recommended for aerobic sludge digestion at a digester temperature of 15°C is:

- (A) 20 days
- (B) 40 days
- (C) **60 days** ✓
- (D) 90 days

Answer: (C)

The Handbook aerobic digestion design-criteria table specifies: $\text{SRT} \geq 40$ days at 20°C and $\text{SRT} \geq 60$ days at 15°C. The longer SRT at lower temperature compensates for slower microbial activity. At temperatures below 15°C the SRT requirement increases further and aerobic digestion may become impractical. Option (B) (40 days) is the 20°C requirement.

Handbook: Environmental Engineering — Aerobic Digestion (Design Criteria table) • Ctrl-F: Aerobic Digestion; Sludge Retention Time; SRT

Q20. [D7-Q20 • Numeric • Per capita BOD loading — wastewater characterization (USCS)]

A community of 25{,}000 people generates wastewater at a per-capita rate of 100 gallons per capita per day (gpcd) with a per-capita BOD₅ load of 0.20 lb/(person·day). Using the Handbook unit-conversion factor $M \text{ (lb/day)} = C \text{ (mg/L)} \times Q \text{ (MGD)} \times 8.34$, determine the BOD₅ concentration C in the raw wastewater (mg/L).

$$\text{Relevant: } Q \text{ (MGD)} = \frac{\text{Population} \times \text{flow (gpcd)}}{1,000,000} \quad M \text{ (lb/day)} = \text{Population} \times \text{per-capita BOD (lb/person/day)}$$

$$C = \frac{M}{Q \times 8.34}$$

Numeric entry.

Answer: $\approx 240 \text{ mg/L}$

$Q = 25,000 \times 100/1,000,000 = 2.50 \text{ MGD}$. $M = 25,000 \times 0.20 = 5,000 \text{ lb/day}$. $C = M/(Q \times 8.34) = 5,000/(2.50 \times 8.34) = 5,000/20.85 = 239.8 \approx 240 \text{ mg/L}$. Domestic raw wastewater typically runs 200–300 mg/L BOD₅; this result is characteristic.

Handbook: Environmental Engineering — Mass Balance / Flow Rate ($M = CQ \times 8.34$) • Ctrl-F: lb/day; MGD; 8.34; Per Capita

Q21. [D7-Q21 • MCQ • Water conservation — per-capita demand reduction]

A water-conservation program reduces per-capita water demand from 150 gpcd to 120 gpcd for a service population of 40{,}000 people. The daily water savings in million gallons per day (MGD) is:

Relevant: $\Delta Q = \Delta(\text{gpcd}) \times \text{Population}/1,000,000$

- (A) 0.030 MGD
- (B) 0.30 MGD
- (C) **1.2 MGD** ✓
- (D) 4.8 MGD

Answer: (C)

$\Delta(\text{gpcd}) = 150 - 120 = 30 \text{ gpcd}$. $\Delta Q = 30 \times 40,000/1,000,000 = 1,200,000 \text{ gpd} = 1.2 \text{ MGD}$. Option (A) = 30 gpcd expressed in MGD without multiplying by population. Option (B) is off by a factor of 4 (perhaps divided by 10{,}000 instead of 1{,}000{,}000). This type of calculation is fundamental to planning infrastructure upgrades, demand forecasting, and justifying reuse projects.

Handbook: N/A — Conservation arithmetic; $M = CQ$ relationship (Handbook p. 327) • Ctrl-F: Per Capita; Demand; Conservation